

C653

Heredity and Genetics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A plant breeder at a seed cooperative that has operated since 1985 crosses two pea plants that are both heterozygous for flower color (Pp). Purple (P) is dominant to white (p). The greenhouse runs at 24°C with 14 hours of light. Out of 400 offspring, about how many are expected to have white flowers?

- A. 300
- B. 100
- C. None, because both parents have purple flowers.
- D. 200

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A purple-flowered pea plant of unknown genotype is crossed with a white-flowered plant. The offspring are about half purple and half white. What is the genotype of the purple parent?

- A. PP
- B. It cannot be known without a second generation.
- C. Pp
- D. pp

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

In a teaching garden that a middle school has used for a decade, a teacher crosses two pea plants that are heterozygous for both seed shape (Rr) and seed color (Yy). The genes are on different chromosomes. Students counted 556 seeds last spring. What fraction of offspring is expected to be wrinkled and green (rryy)?

- A. 9/16
- B. 1/16
- C. 1/4
- D. 3/16

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A plant that is AaBb is crossed with one that is aabb. The genes assort independently. What fraction of offspring will be Aabb?

- A.** 1/16
- B.** 1/2
- C.** 1
- D.** 1/4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A rancher breeds a red bull with a white cow. The calf, born during an unusually cold spring, has a coat with red hairs and white hairs mixed evenly all over its body. The rancher has kept this herd for three generations. Which inheritance pattern best explains the calf's coat?

- A.** Codominance
- B.** Complete dominance of red over white
- C.** Incomplete dominance
- D.** A temperature effect caused by the cold spring

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A woman who is a carrier for red-green color blindness, an X-linked recessive trait, has children with a man who has normal color vision. Her brother is color-blind and works as a graphic designer. What is the probability that any one of their sons will be color-blind?

- A.** 0
- B.** 1/4
- C.** 1, because boys always inherit X-linked traits from their mothers.
- D.** 1/2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A man with hemophilia (X-linked recessive) has children with a woman who has two normal alleles. What fraction of their daughters will be carriers?

- A.** All of them
- B.** None
- C.** 1/2
- D.** 1/4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A lab using fruit flies ordered from a national stock center does a test cross for two linked genes. The 1,000 offspring are: 420 and 410 in the parental classes, and 90 and 80 in the recombinant classes. The lab keeps the vials at 25°C. What is the map distance between the genes?

- A.** 17 map units
- B.** The genes are unlinked, since any recombinants prove it.
- C.** 83 map units
- D.** 9 map units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A field team surveys a beetle population of 2,000 on a remote island that has been a nature preserve since 1970. Black body color is recessive (bb), and 320 beetles are black. Assuming Hardy-Weinberg equilibrium, what is the frequency of the b allele?

- A.** 0.84
- B.** 0.4
- C.** 0.16
- D.** 0.6

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

In a population in Hardy-Weinberg equilibrium, 9% of individuals show a recessive trait. What fraction are heterozygous carriers?

- A. 0.49
- B. 0.21
- C. 0.09
- D. 0.42

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A student in a biotech elective reads a short DNA template strand from a class worksheet that took two weeks to design: 3'-TAC GGA-5'. What mRNA sequence is transcribed from it?

- A. Methionine-Proline
- B. 5'-ATG CCT-3'
- C. 5'-AUG CCU-3'
- D. 5'-UAC GGA-3'

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A gene has 300 codons and was first sequenced by a university lab in 2004. Which change is most likely to produce a completely nonfunctional protein? The protein is found in high amounts in liver cells.

- A. A deletion of three bases in codon 2
- B. A substitution in the third base of codon 150
- C. A single-base insertion in codon 2
- D. Any of these, since every mutation destroys the protein.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A plant breeder ~~at a seed cooperative that has operated since 1985~~ crosses two pea plants that are both heterozygous for flower color (Pp). Purple (P) is dominant to white (p). ~~The greenhouse runs at 24°C with 14 hours of light.~~ Out of 400 offspring, about how many are expected to have white flowers?

Target word: expected number of recessive offspring **Decides:** Pp × Pp gives 1/4 pp.

A	Inverted operation	That is the number expected to show purple, the dominant phenotype.
B	Correct	1/4 of 400 offspring are pp and show white flowers.
C	Absolute language	Heterozygous parents each carry p and can pass it on together.
D	Confusion set	That is the expected number of heterozygotes (Pp), not white-flowered plants.

2. Answer C Concept 1 · transfer

A purple-flowered pea plant of unknown genotype is crossed with a white-flowered plant. The offspring are about half purple and half white. What is the genotype of the purple parent?

Target word: solve backward from offspring ratio to parent genotype **Decides:** A 1:1 ratio from a test cross means the unknown parent is heterozygous.

A	Inverted operation	PP × pp would give all purple offspring.
B	Dropped element	Ignores the 1:1 ratio, which already settles the genotype.
C	Correct	Pp × pp gives 1/2 Pp (purple) and 1/2 pp (white).
D	Confusion set	pp is white, but the parent in question has purple flowers.

3. Answer B Concept 2

In a teaching garden ~~that a middle school has used for a decade~~, a teacher crosses two pea plants that are heterozygous for both seed shape (Rr) and seed color (Yy). The genes are on different chromosomes. ~~Students counted 556 seeds last spring.~~ What fraction of offspring is expected to be wrinkled and green (rryy)?

Target word: probability of the double-recessive combination **Decides:** Independent genes: 1/4 rr × 1/4 yy.

A	Inverted operation	That is the share showing both dominant traits.
B	Correct	$1/4 \times 1/4 = 1/16$ for two independent genes.
C	Dropped element	Uses only one gene's probability.
D	Adjacent category	That is the share with one dominant and one specific recessive trait.

4. Answer D Concept 2 · transfer

A plant that is AaBb is crossed with one that is aabb. The genes assort independently. What fraction of offspring will be Aabb?

Target word: product rule in a dihybrid test cross **Decides:** P(Aa) = 1/2 and P(bb) = 1/2.

A	Confusion set	Borrows the AaBb × AaBb ratio for a different cross.
B	Dropped element	Accounts for only one of the two genes.
C	Inverted operation	Adds the two probabilities instead of multiplying them.
D	Correct	$1/2 \times 1/2 = 1/4$.

5. Answer A Concept 3

A rancher breeds a red bull with a white cow. The calf, ~~born during an unusually cold spring,~~ has a coat with red hairs and white hairs mixed evenly all over its body. ~~The rancher has kept this herd for three generations.~~ Which inheritance pattern best explains the calf's coat?

Target word: name the pattern **Decides:** Both colors appear fully and separately, not blended.

A	Correct	Each allele is fully expressed in separate hairs.
B	Adjacent category	The calf would then look solid red.
C	Confusion set	Would produce a single blended, in-between hair color.
D	Camouflage	Answers the weather detail, which does not decide the pattern.

6. Answer D Concept 4

A woman who is a carrier for red-green color blindness, an X-linked recessive trait, has children with a man who has normal color vision. ~~Her brother is color blind and works as a graphic designer.~~ What is the probability that any one of their sons will be color-blind?

Target word: probability for a son **Decides:** A son's only X comes from the carrier mother.

A	Inverted operation	That is the chance for a daughter, who also gets a normal X from her father.
B	Confusion set	That is the chance that any child (son or daughter) is color-blind.
C	Absolute language	The mother has two different X chromosomes; sons get only one.
D	Correct	The mother passes either her normal X or her affected X, each with 1/2 chance.

7. Answer A Concept 4 · transfer

A man with hemophilia (X-linked recessive) has children with a woman who has two normal alleles. What fraction of their daughters will be carriers?

Target word: carrier status of daughters **Decides:** Every daughter receives the father's only X.

A	Correct	Each daughter gets his affected X plus a normal X from the mother.
B	Inverted operation	Treats the father as passing his X to sons instead of daughters.
C	Confusion set	Applies an autosomal carrier ratio to an X-linked trait.
D	Dropped element	Ignores that the father has only one X to give daughters.

8. Answer A Concept 5

A lab using fruit flies ~~ordered from a national stock center~~ does a test cross for two linked genes. The 1,000 offspring are: 420 and 410 in the parental classes, and 90 and 80 in the recombinant classes. ~~The lab keeps the vials at 25°C.~~ What is the map distance between the genes?

Target word: map distance **Decides:** Recombinants = 90 + 80 = 170 of 1,000.

A	Correct	$170 / 1,000 = 17\%$ recombination = 17 map units.
B	Absolute language	Linked genes still recombine by crossing over; only about 50% recombination signals no linkage.
C	Inverted operation	Uses the parental classes (830) instead of the recombinants.
D	Dropped element	Counts only one of the two recombinant classes.

9. Answer B Concept 6

A field team surveys a beetle population of 2,000 ~~on a remote island that has been a nature preserve since 1970~~. Black body color is recessive (bb), and 320 beetles are black. Assuming Hardy-Weinberg equilibrium, what is the frequency of the b allele?

Target word: recessive allele frequency **Decides:** $q^2 = 320 / 2,000 = 0.16$.

A	Confusion set	That is the share of beetles showing the dominant phenotype.
B	Correct	$q = \sqrt{0.16} = 0.4$.
C	Dropped element	That is q^2 , the genotype frequency; the square root is missing.
D	Inverted operation	That is p, the dominant allele frequency.

10. Answer D Concept 6 · transfer

In a population in Hardy-Weinberg equilibrium, 9% of individuals show a recessive trait. What fraction are heterozygous carriers?

Target word: solve for 2pq **Decides:** $q^2 = 0.09$, so $q = 0.3$ and $p = 0.7$.

A	Confusion set	That is p^2 , the homozygous dominant share.
B	Dropped element	pq without the factor of 2.
C	Wrong stem number	The recessive phenotype share given in the stem.
D	Correct	$2pq = 2 \times 0.7 \times 0.3 = 0.42$.

11. Answer C Concept 7

A student ~~in a biotech elective~~ reads a short DNA template strand from a class worksheet ~~that took two weeks to design~~: 3'-TAC GGA-5'. What mRNA sequence is transcribed from it?

Target word: mRNA sequence **Decides:** mRNA pairs with the template and uses U for A.

A	Confusion set	That is the protein product of translation, not the mRNA.
B	Adjacent category	That is the coding DNA strand; RNA uses U, not T.
C	Correct	T→A, A→U, C→G, G→C, running antiparallel.
D	Inverted operation	Copies the template instead of pairing with it.

12. Answer C Concept 8

A gene has 300 codons ~~and was first sequenced by a university lab in 2004~~. Which change is most likely to produce a completely nonfunctional protein? ~~The protein is found in high amounts in liver cells.~~

Target word: most damaging mutation **Decides:** A one-base insertion early shifts the reading frame for all later codons.

A	Adjacent category	Removes one amino acid but keeps the reading frame.
B	Confusion set	Third-position changes are often silent due to the redundant code.
C	Correct	Frameshift near the start scrambles nearly every downstream codon.
D	Absolute language	Many mutations are silent or mild.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Monohybrid cross ratios	A cross of two heterozygotes ($Aa \times Aa$) gives a 1:2:1 genotype ratio and a 3:1 dominant-to-recessive phenotype ratio.	Assuming two parents who show the dominant trait cannot have offspring with the recessive trait.	Both parents are heterozygous, or a 3:1 / 1:1 offspring count is given.
2	Independent assortment and the product rule	For unlinked genes, the probability of a combined outcome is the product of each gene's separate probability.	Adding probabilities, or using 9:3:3:1 for crosses that are not $AaBb \times AaBb$.	Two genes and a question about one specific combination.
3	Incomplete dominance vs. codominance	Incomplete dominance blends into an intermediate phenotype; codominance shows both alleles fully and separately.	Calling any heterozygote that looks different from both parents "incomplete dominance."	An intermediate color (blend) vs. patches or spots of both colors.
4	X-linked recessive inheritance	Sons get their only X from the mother; daughters get one X from each parent, so fathers pass X-linked alleles to all daughters and no sons.	Treating X-linked traits like autosomal ones, or thinking fathers pass X alleles to sons.	A trait seen mostly in males, or a "carrier" mother.
5	Linkage and map distance	Recombination frequency = recombinant offspring / total offspring; 1% recombination = 1 map unit.	Using the parental (most common) classes as the recombinants.	Four offspring classes in two large and two small counts.
6	Hardy-Weinberg equilibrium	$p + q = 1$ and $p^2 + 2pq + q^2 = 1$; the recessive phenotype frequency is q^2 , so q is its square root.	Reading the recessive phenotype frequency as q without taking the square root.	The share of individuals showing a recessive trait in a large, randomly mating population.
7	Transcription base pairing	mRNA is built complementary and antiparallel to the DNA template strand, with U in place of T.	Copying the template strand directly or keeping T in the RNA.	A template-strand sequence and a request for the mRNA.
8	Frameshift vs. point mutations	Inserting or deleting bases in numbers other than multiples of three shifts the reading frame and changes every codon downstream.	Assuming every single-base change has the same effect, or that all mutations are harmful.	A choice between an insertion/deletion and a substitution.

Course scope (public description): Heredity and Genetics is an introductory course for graduate students seeking initial licensure or endorsement and/or students earning their MA degree in secondary or middle grade science education. This course addresses the basic principles of heredity and the function of molecular genetics. Topics include Mendelian and non-Mendelian inheritance and population genetics. This course has no prerequisites.